

Ajay Suresh

 [ajay-suresh-ugz186](#) |  ajay.suresh@my.utsa.edu |  [ajaysuresh24](#)

Education

Doctor of Philosophy in Mechanical Engineering, University of Texas at San Antonio, TX, USA Jun '26 – May '29
Master of Science in Aerospace Engineering, University of Michigan, Ann Arbor, MI, USA Aug '24 – May '26
Bachelor of Science in Aerospace Engineering, Madras Institute of Technology, Chennai, India Aug '19 – May '23

Skills

1. Siemens NX | 2. SolidWorks | 3. OpenFOAM | 4. MSC Nastran | 5. Python | 6. MATLAB
| 7. Weather Research and Forecasting (WRF) | 8. WRF-Chem | 9. Remote Sensing (Satellite Data Analysis, GIS) | 10. ArcGIS | 11. High-Performance Computing (MPI, SLURM, Parallel Computing)

Research and Professional Experience

Graduate Research Assistant | NASA-Funded ISRAP-AIRMAPS Project Jun '26 - Present

Developing an integrated urban air-quality modeling framework for San Antonio using WRF/WRF-Chem, mobile measurements, remote sensing, and meteorological observations. Conduct plume-transport simulations and validate model performance against observational datasets to improve characterization of pollutant transport and dispersion.

Research Intern | DRDO - Supersonic UAV Intake Design Jan '23 - Jun '23

Developed a supersonic UAV S-duct intake in Siemens NX using Gerlach shaping to improve flow quality from the intake throat to the Aerodynamic Interface Plane (AIP), then generated a hybrid computational mesh in ANSYS ICEM CFD for subsequent RANS analysis to evaluate pressure loss and secondary-flow development, leading to geometry refinements that reduced flow separation by approximately 5% and improved downstream flow uniformity.

Academic Projects

Numerical Modeling of Transatlantic Saharan Dust Transport Using WRF-Chem Jun '26 – Present

Investigating Saharan dust transport from North Africa across the Atlantic to Central Texas using WRF-Chem with ERA5 meteorological forcing. Analyzing size-resolved dust concentrations, aerosol optical depth, wind fields, and boundary-layer dynamics to characterize plume evolution and vertical dust distribution.

Numerical Investigation of 2D Airfoil Flow Using RANS Aug '25 - Dec '25

Developed a Python-based 2D airfoil CFD workflow with near-wall mesh refinement, boundary-condition setup, and SST $k - \omega$ RANS turbulence modeling. Performed mesh-sensitivity analysis and validated aerodynamic predictions against established reference data, with further analysis of pressure distribution, flow separation, and downstream turbulent wake development.

Thermodynamic Modeling and Surrogate Prediction of Turbojet Engine Jan '25 - May '25

Developed an integrated turbojet model treating the compressor, combustor, turbine, and nozzle as coupled black-box subsystems to study how component interactions affect overall engine performance. Built a surrogate prediction model to approximate thrust and cycle performance across variations in pressure ratio, mass flow, and thermal limits, achieving predictions within approximately 5% of the reference cycle model.

Turbojet Multidisciplinary Design Optimization (MDO) Aug '24 - Dec '24

Developed a real-gas turbojet performance model using Cantera to evaluate thermodynamic properties and component-level cycle behavior. Applied a Genetic Algorithm for global design-space exploration followed by Nelder-Mead local refinement to optimize thrust, specific fuel consumption (SFC), and thermal constraints, improving net thrust by approximately 7% and reducing design iteration time by approximately 25%.

Positions of Responsibility

Student Mentor | NASA-Funded ISRAP-AIRMAPS Project Jun '26 - Present

Mentor an interdisciplinary group of undergraduate students from multiple departments, providing guidance on research activities, technical workflows, and project coordination within the NASA-funded ISRAP-AIRMAPS program.

Student Mentor | Raptors Flight Club, Madras Institute of Technology May '21 - May '23

Mentored and onboarded new team members, providing guidance on aircraft-design activities, team responsibilities, and project workflows. Organized a national-level Aircraft Design workshop as part of the *Flight* symposium.